

CoolDesk

AC Service Management Portal **Architecture Document**

Version 5.2 | 19 Modules | 4 User Roles | AWS / NestJS / Next.js / PostgreSQL

1. System Overview

CoolDesk is a role-based operations portal for managing AC installations and complaints. It replaces manual tracking via WhatsApp, paper, and spreadsheets with a structured, state-driven workflow that provides end-to-end visibility across jobs, technicians, brands, and the dealer network.

All work in the system is represented as a job. Every job must follow a strict, forward-only lifecycle enforced at the service layer. The system is not CRUD-based; it is state-driven. The owner holds master credentials and is responsible for creating and managing all other user accounts.

2. Core Data Model

2.1 Jobs

A job is the central entity. Every installation and every complaint is stored in the same table and distinguished only by a type field. A job also carries a source field (Direct or Via Dealer) that determines how it entered the system. All jobs share a uniform operational lifecycle; source only affects filtering, analytics, and display tagging.

No job can exist without all of the following fields: type, status, brand, and source. A technician assignment is not required at the moment of job creation but is required before a job can advance beyond its initial state. Complaint-specific fields must not appear on installation jobs and vice versa.

2.2 Customer Data and Virtual Customer Identity

Customer data is embedded directly inside each job. There is no separate customer table. Each job stores its own customer name, multiple phone numbers, and address.

To maintain accurate identity linkage across multiple jobs, the system assigns a Virtual Customer ID (VCID) to each unique customer. The VCID is generated automatically and mapped to the primary phone number at the time of first job creation. It is a backend construct only — the VCID identifier itself is never displayed or referenced by name in the UI. All linking actions are presented to users in terms of customer identity — name, address, and phone numbers. The VCID resolves and updates in the backend transparently as a result of these actions. It is the sole identifier used by the repeat complaint and frequent complaint detection logic.

When logging a new job, the system checks the entered phone number against existing VCIDs and prompts the user with a 'Link to Existing Customer' suggestion if a match is found, displaying the matched customer's name and address for confirmation. The prompt also fires on a name and address partial match, covering cases where a returning customer provides a different phone number. Additional phone numbers entered on subsequent jobs can be manually linked to an existing VCID by office staff during the lookup flow — presented as linking to the matched customer's profile, with the VCID resolving in the backend.

VCID linking is a soft recommendation, not a hard gate. The user may dismiss the linking prompt and proceed, in which case a new VCID is generated automatically for that job. A dismissed linking prompt is logged silently as "Customer link suggestion dismissed — [matched customer name] — [actor] —

[timestamp]", allowing office staff or the owner to review and manually link records later if needed. Every job always carries a VCID — either linked to an existing one or newly generated.

The VCID is the authoritative link between a customer and all their historical jobs. Without it, a customer calling from a new number would appear as a new customer and bypass repeat complaint and frequent complaint detection entirely.

2.3 Customer Lookup

Customer lookup is available in the office staff portal and supports search by both name and phone number. When multiple customers share the same name, results display the customer name, all linked phone numbers, and address together so staff can identify the correct entry at a glance. The most recent job per customer is shown alongside for additional context. On a confirmed match, the job form is pre-filled with the customer's details and linked to their existing VCID.

2.4 Brands

Every job must be associated with a brand. Brands are managed by the owner and can be activated or deactivated but never deleted. Brand association enables filtering across all operational views and powers brand-level analytics including revenue, complaint rates, and revisit rates. Brands can be colour-coded for visual identification.

2.5 Payments

Every job has exactly one payment record. Payment is recorded at the end of the job lifecycle and includes amount, payment method, and status (Collected or Pending). Technicians are responsible for recording payment at the point of job completion.

Payment authority is split by role. Technicians record the initial payment. Office staff may modify the Payment Method field only (for example, to correct a data entry error). Only the owner may modify the Payment Amount. Any edit to a payment record by any role is logged in the job timeline with the previous value, the new value, the timestamp, and the actor who made the change.

The list of accepted payment methods (e.g. Cash, Bank Transfer, Cheque) is defined and managed by the owner in system settings. This list is stored centrally and appears as a dropdown when technicians record payment. The owner can add or deactivate payment methods at any time, ensuring consistent data across all records. Payment data feeds into analytics and outstanding balance tracking. Payment is brand-tagged, enabling revenue reporting per brand.

When the owner reverses a Completed or Resolved status outside the 60-second undo window, and the job has a recorded payment, the system presents a mandatory decision prompt requiring the owner to either retain or void the payment record. Retaining keeps the payment as-is, marked as "Recorded on prior completion." Voiding reverses the payment record; the owner must provide a reason, which is logged in the job timeline alongside the previous values, timestamp, and actor. Neither action is taken automatically and the prompt cannot be bypassed.

2.6 Repeat Complaint Detection

Repeat complaint detection compares new complaint entries against the full job history of the same VCID using configurable thresholds. The default repeat window is 30 days. Customers with three or more complaints linked to the same VCID in any 90-day window are flagged as frequent complaint customers.

Because detection operates on the VCID rather than raw phone number, it remains accurate across multiple contact numbers and across both direct and dealer-submitted jobs.

2.7 Multi-Unit Jobs

A single job is tied to one customer address. For installations or complaints involving multiple AC units at the same address, the job supports multiple unit entries within a single record — each with its own model, unit type, and fault or installation description. If a customer has units at separate locations, those are logged as separate jobs and linked to the same customer VCID via phone number lookup.

3. User Roles and Authentication

There are four user roles. The owner holds master credentials and is the only account that exists independently. All other accounts — office staff, technicians, and dealers — are created and managed by the owner. Owner, Office Staff, and Technician exist in the main users table and are authenticated through the core system. Dealers are a separate entity with their own login credentials issued by the owner.

Role	Capabilities
Owner	Full read-write access. Can override any record, status, payment amount, or payment method. Exclusively authorised to reverse Completed and Resolved statuses and reopen cancelled jobs. Creates and manages all user accounts (technicians, office staff, dealers). Manages brands, payment methods, and all system settings. Receives dealer cancellation request notifications alongside office staff.
Office Staff	Creates jobs (installations and complaints), assigns and reassigns technicians, manages revisit scheduling, performs customer lookup by name or phone number, and actions dealer submissions. Can roll a job back by exactly one state (pre-completion and pre-resolution only) with a mandatory reason logged. Can modify Payment Method (not Amount). Can approve or reject dealer cancellation requests. Sees scheduling conflict warnings and amber revisit-pending alerts.
Technician	Interacts only with assigned jobs. Moves jobs forward through their lifecycle, logs arrival, adds notes, records payment (method and amount), and copies customer phone number. Has a 60-second undo window after each status transition. Cannot assign or reassign jobs. Cannot modify payment once recorded.
Dealer	Separate entity with own login (issued by owner). Submits complaint and installation jobs on behalf of customers. Can directly withdraw a job only while it is in its initial state (New for complaints, Pending Schedule for installations). For jobs beyond the initial state, can submit a Request Cancellation that requires office staff or owner approval. Views the status and full history of their own submissions in read-only format, including cancellation request outcomes. Cannot access other dealers' data or assign technicians.

Role-based guards are enforced on every API request via JWT tokens so that users can only access what they are permitted to see or modify.

4. Job Lifecycles

Every job moves through a predefined sequence of statuses. Transitions are strictly enforced in the service layer. A job cannot skip states, move backward, or enter an invalid state for its type. Technicians can only move a job forward and cannot edit a status once the undo window has elapsed. Office staff can roll a job back by exactly one state (pre-completion and pre-resolution only). The owner can override job status at any stage.

4.1 Installation Status Flow

Every installation job moves through a six-stage track from booking to completion, with Cancelled as an additional terminal state.

Label	Status	Meaning
PS	Pending Schedule	Dealer-submitted installation job awaiting date assignment by office staff. Not applicable to direct installation jobs.
S	Scheduled	Job booked and logged. Date and time set.
A	Assigned	Technician assigned. In-app notification sent.
P	Acknowledged	Technician confirmed receipt. Timestamp recorded.
T	In Transit	Technician is en route to the site.
W	In Process	Technician is on site, performing installation.
C	Completed	Installation done. Payment recorded.
X	Cancelled	Job cancelled. Reason logged. No further action taken.

When a dealer submits an installation job, it enters Pending Schedule before the standard lifecycle begins. Office staff see all Pending Schedule jobs in a dedicated queue on their portal. Once a date and time are set, the job transitions to Scheduled and follows the standard installation lifecycle. Dealers see this state displayed as "Awaiting Scheduling" in their history view. The owner receives a notification on each dealer installation submission linking directly to the job in Pending Schedule.

PAYMENT AND CONNECTIVITY

The Completed (and Resolved) status transition and payment recording are treated as a single atomic operation. Both are queued together in the sync queue and submitted to the server as one entry when connectivity is restored. The undo window begins immediately on the technician's device at the moment of submission to the queue — not upon server confirmation — so the technician can undo without waiting for a sync. The server-side lock is applied once the sync lands.

4.2 Complaint Status Flow

Every complaint follows the primary track from logging to resolution, regardless of source. Cancelled and Cancellation Requested are additional states available within the lifecycle.

Label	Status	Meaning
N	New	Complaint logged (direct or via dealer). Awaiting assignment.
A	Assigned	Technician selected. Notification sent.
P	Acknowledged	Technician confirmed receipt.
T	In Transit	Technician is en route.
W	In Process	Technician on site, working on issue.
R	Resolved	Issue fixed. Payment recorded. Complaint closed.
V	Needs Revisit	Issue unresolved. Reason logged. Revisit record created. Amber alert raised.
S	Revisit Scheduled	Return visit date and time set. Amber alert cleared.
RV	Resolved on Revisit	Issue fixed on return visit. Payment recorded. Complaint closed.
CR	Cancellation Requested	Dealer has requested cancellation. Awaiting office staff or owner approval.
X	Cancelled	Job cancelled or withdrawn. Reason logged. No further action taken.

A complaint can only be fully closed when a technician explicitly marks it as Resolved or Resolved on Revisit. Both are equivalent terminal states for the purposes of analytics, resolution rate calculations, and payment recording. This applies equally to direct complaints and dealer-submitted complaints.

4.3 Graceful Undo Window

Technicians have a 60-second undo window immediately after any status transition to revert an accidental tap. The window begins on the technician's device at the moment of submission — for queued entries, it does not wait for server confirmation. A visible countdown is shown on screen alongside a single Undo button. After 60 seconds the transition is locked on the device and cannot be reversed by the technician — only office staff (one-state rollback) or the owner (full override) can intervene. The server-side lock is enforced once the queued entry syncs.

Two transitions carry additional rollback logic:

- Needs Revisit — if undone within the window, the newly created revisit record is deleted and the amber pending scheduling alert is automatically cleared. The complaint returns to its previous In Process status.
- Completed / Resolved — if undone within the window, the payment entry submitted as part of the same atomic sync entry is also rolled back in full. The job returns to In Process.

SYSTEM CONSTANT

The undo window duration is set to 60 seconds. This is a system-level constant, not currently configurable per client. If per-client configurability is required in future, this value should be moved into the Configuration Layer.

4.4 Office Staff One-State Rollback

Office staff can move a job back by exactly one state to correct an error made by a technician or during job entry. The following rules govern this capability:

- Rollback is restricted to pre-completion and pre-resolution stages only. The Completed and Resolved statuses are owner-only reversals.
- A mandatory reason must be provided before the rollback is applied.
- The rollback event is logged in the job timeline with the reason, the previous status, the reverted status, the timestamp, and the office staff member who performed it.
- Rollback is not available on Cancelled jobs. A cancelled job can only be reopened by the owner.

4.5 Revisit Track

Triggered when a technician marks a complaint as Needs Revisit. A revisit record is automatically created and linked to the parent complaint. Revisits are not independent jobs; they are child records representing follow-up visits. The parent complaint remains open until a final resolution occurs.

Each revisit has its own schedule, technician assignment, arrival tracking, and outcome. A revisit can either resolve the issue or trigger another revisit, forming a loop until resolution.

Before marking Needs Revisit, the technician must select one of five mandatory reasons:

- Part unavailable — required part not on hand
- Customer not home — could not access premises
- Issue recurring — problem came back after a previous fix
- Further diagnosis required — needs specialist review
- Custom note — technician enters their own reason

Each revisit is assigned to a technician independently, though by default it inherits the technician from the parent job. A different technician can be assigned if required.

When a third revisit record is created against the same parent complaint — meaning Needs Revisit has been triggered three times, resulting in a total of four site visits including the original — the owner and office staff are notified via the notification centre. The job is flagged as a chronic unresolved complaint and receives a distinct visual indicator on the owner dashboard, separate from the standard amber revisit-pending alert. This indicator persists until the complaint reaches a final resolution status.

REVISIT PENDING SCHEDULING ALERT

When a technician marks a complaint as Needs Revisit, a distinctly coloured amber alert card is immediately raised on both the office staff portal and the owner dashboard. This alert remains visible and prominent until office staff set a revisit date and time, at which point it is automatically cleared. If the Needs Revisit transition is undone within the 60-second undo window, the alert is cleared immediately without requiring office staff action.

4.6 Cancellation and Withdrawal Flow

A Cancelled terminal status is available on both installation and complaint jobs. The following rules apply:

- For direct jobs: office staff or the owner can cancel at any stage before the job is completed or resolved. A mandatory cancellation reason must be provided.
- For dealer-submitted complaint jobs in New status: the dealer can withdraw the job directly. A mandatory withdrawal reason must be provided. For dealer-submitted installation jobs, Pending Schedule is treated as the withdrawal-eligible initial state, equivalent to New for complaint jobs.

The dealer may withdraw the job directly while it remains in Pending Schedule. Once the job advances to Scheduled, the standard cancellation request process applies.

- For dealer-submitted jobs in any status beyond their initial state (i.e. already assigned or scheduled): the dealer cannot cancel directly. Instead, a Request Cancellation button is available in the dealer portal. Submitting a request transitions the job to Cancellation Requested status and raises a notification to both office staff and the owner.
- While in Cancellation Requested status, a visible badge appears on the job in the office staff and owner views. The job continues its normal lifecycle display but is flagged for decision.
- Office staff or the owner can approve or reject the request in one click. If approved, the job moves to Cancelled with the dealer's stated reason logged. If rejected, the job returns to its previous status and continues normally.
- The dealer sees the outcome of their request (Approved or Rejected) in their history view. A dealer cannot submit another cancellation request for the same job once one has been rejected.
- Cancelled jobs are excluded from resolution rate and completion rate analytics but included in total job volume counts.
- The owner can reopen a cancelled job, returning it to its previous status. All cancellation and reopening events are logged in the job timeline with reason, actor, and timestamp.

4.7 Visit Attendance Outcomes

Punctuality is derived by comparing the scheduled visit time against the actual logged arrival time, using a configurable grace period (default: 15 minutes). Every scheduled visit — first visit, revisit, or installation — receives one of the following outcomes:

Outcome	Description
On Time	Arrived within the configurable grace period (default: 15 minutes).
Late	Arrived after grace period. Delay duration recorded in minutes.
No-Show	No arrival recorded. Job flagged on owner dashboard for follow-up.
Rescheduled	Visit moved before it occurred. New slot assigned.

No-Show visits are automatically flagged on the owner dashboard. Unacknowledged jobs are also surfaced to the owner and office staff through dashboard alerts.

5. Technician Assignment

Once a job has been assigned, it must always have exactly one active technician at any given time. Assignment is controlled by office staff or the owner. Technicians cannot assign or reassign jobs.

When a job is assigned, the system records the assignment timestamp and logs the event in the assignment history table. If a job is reassigned, the new assignment overwrites the active technician while preserving all previous assignments in history.

When a job is assigned, the technician must acknowledge it. Acknowledgment is mandatory and timestamped. The time between assignment and acknowledgment is tracked and used in performance analytics. Jobs not acknowledged within a reasonable window are surfaced to the owner and office staff on the dashboard.

Revisit assignments are independent of the parent job assignment. By default the revisit inherits the parent job technician, but a different technician can be assigned if required.

5.1 Scheduling Conflict Detection

When office staff or the owner assign a technician to a job, the system checks whether that technician already has another job scheduled in an overlapping time window. If a conflict is detected, a warning is surfaced on the assignment screen before the assignment is confirmed.

The conflict warning is advisory, not a hard block — the assignment can still proceed, as office staff may have context the system does not. However, the conflict must be explicitly acknowledged before continuing. This applies to both first-visit assignments and revisit assignments.

When a conflict warning is acknowledged and the assignment proceeds, this action is logged in the job timeline as: "Scheduling conflict acknowledged by [actor] on [timestamp]. Conflicting job: [Job ID]. Assignment proceeded." This ensures a clear audit record exists if the double-booking causes an issue later.

6. Dealer Network

Dealers are a separate entity layer within the system. Each dealer has their own login credentials (issued by the owner), business details, operational region, and associated brands. Dealers can be activated or deactivated at any time without deleting their historical data.

6.1 Dealer Job Submission

Dealers can submit both complaint jobs and installation jobs on behalf of their customers. The dealer portal submission form includes a job type selector — Complaint or Installation — and the form adapts accordingly.

Complaint submissions capture: customer name, phone number, address, AC brand, model, and issue description.

Installation submissions capture: customer name, phone number, address, AC brand, model, unit type, and number of units — mirroring what office staff enter directly.

Once submitted, the job enters the standard lifecycle for its type — complaints start at New, dealer-submitted installations enter Pending Schedule until office staff set a date — and is tagged with its dealer source. The owner receives an in-app notification on each dealer submission. From that point, the job follows the identical lifecycle to a direct job with no special-case workflow logic. Phone numbers entered on dealer submissions are checked against existing VCIDs and linked automatically or flagged for manual linking by office staff.

6.2 Dealer Job History and Status View

Dealers have a read-only history view within their portal showing all complaints and installation requests they have ever submitted, along with the current status of each. Dealer-submitted installations in Pending Schedule are shown with the display label "Awaiting Scheduling." This view is filterable by job type, date range, and status. It also displays the outcome of any cancellation requests the dealer has submitted (Approved or Rejected).

Dealers cannot edit any record. The view is purely informational, allowing dealers to respond accurately to their customers regarding job progress without requiring a phone call to the office.

Dealers cannot view other dealers' submissions, assign technicians, or access any analytics or internal operational data.

7. Customer Contact in the Field

The technician mobile view displays full customer details on each job card, including customer name, address, and all linked phone numbers. A copy button is available next to each phone number, allowing the technician to copy it to their clipboard and paste it directly into their phone's dialler. No tap-to-call is built into the system; the copy mechanism keeps the interaction simple and device-agnostic.

7A. Notification Centre

Notifications are delivered exclusively through an in-app notification centre. There are no push notifications. The notification centre is accessible from a persistent icon in the navigation bar across all role portals. Unread notifications are indicated by a badge count on the icon. Clicking a notification navigates the user directly to the relevant job. All notifications are stored persistently with a read/unread state.

Each role receives the following notifications:

Event	Owner	Office Staff	Technician	Dealer
New dealer job submitted	Yes	Yes	—	—
Job assigned to technician	—	—	Yes	—
Dealer cancellation request submitted	Yes	Yes	—	—
Cancellation request outcome (approved/rejected)	—	—	—	Yes
Job unacknowledged beyond threshold	Yes	Yes	—	—
No-show flagged	Yes	Yes	—	—
Repeat or frequent complaint detected	Yes	Yes	—	—
Complaint reaches third revisit	Yes	Yes	—	—

8. Module Structure

CoolDesk is organised into 19 modules across six categories. Module 18 covers both complaint and installation submission from the dealer portal. Module 19 is an optional add-on and can be included or excluded without affecting any of the 18 core modules.

#	Module	Category	Type
01	AC Installation Tracking	Core Operations	Core
02	Complaint Management	Core Operations	Core
03	Technician Assignment	Core Operations	Core
04	Notifications and Acknowledgment	Core Operations	Core
05	Live Status Updates from Field	Core Operations	Core
06	Payment Tracking	Core Operations	Core
07	Revisit Scheduling and Tracking	Revisit & Scheduling	Core
08	Punctuality Tracking	Revisit & Scheduling	Core
09	Owner Full Read-Write Access	Owner Portal & Alerts	Core
10	Owner Dashboard	Owner Portal & Alerts	Core
11	Repeat Complaint Alert System	Owner Portal & Alerts	Core
12	Office Staff Portal	Staff Portals	Core
13	Technician Mobile View	Staff Portals	Core
14	Brand / Company Segregation	Brand Management	Core
15	Technician Performance Analytics	Analytics	Core
16	Business-Wide + Dealer Analytics	Analytics	Core
17	Dealer Management	Dealer Network	Core
18	Dealer Job Submission	Dealer Network	Core
19	Customer Reviews and Ratings	Optional Add-on	Optional

8.1 Category A — Core Operations (Modules 01–06)

Covers AC installation tracking, complaint management, technician assignment, in-app notifications with mandatory acknowledgment, live field status updates, and payment tracking. Cancellation, undo window, and split payment authority are all part of each job type's lifecycle within this category.

8.2 Category B — Revisit and Scheduling (Modules 07–08)

Module 07 manages the full revisit lifecycle including mandatory reason selection, scheduling, technician notification, arrival tracking, and outcome recording. The amber revisit-pending alert is raised and cleared within this module, including automatic clearing when the Needs Revisit undo window is used. Module 08 handles punctuality tracking across all visit types using configurable grace periods, and feeds data into analytics.

8.3 Category C — Owner Portal and Alerts (Modules 09–11)

The owner has complete read-write access and can override any record, status, or payment. The owner dashboard provides a unified command centre with live counts, per-technician and per-dealer breakdowns, standard alert queues, amber revisit-pending alert cards, and dealer cancellation request notifications. The repeat complaint alert system auto-detects returning customers by VCID and flags frequent complaint customers. Chronic unresolved complaints — those where Needs Revisit has been triggered three times — are escalated to both the owner and office staff via the notification centre and surfaced with a distinct visual indicator on the owner dashboard.

8.4 Category D — Staff Portals (Modules 12–13)

The office staff portal provides a quick-entry form for logging any job type, customer lookup by name or phone number with VCID-based linking, address-based disambiguation and pre-fill, a unified job list (direct and dealer-submitted together, source-tagged), a dedicated Pending Schedule queue for dealer-submitted installation jobs awaiting scheduling, technician workload visibility, scheduling conflict warnings on assignment, one-state rollback capability with mandatory reason logging, dealer cancellation request approval, revisit slot management, and amber revisit-pending alert cards.

The technician mobile view is optimised for field use on low-end Android devices and basic browsers. It shows assigned jobs with type and source badges, full job detail including customer contact with a phone number copy button, status update controls with the 60-second undo window displayed after each transition, and payment recording. Status transitions and payment details are queued together atomically in the browser sync queue and submitted to the server as a single entry when connectivity is restored. The undo window begins immediately on the device at the moment of queue submission. A subtle indicator is shown if an entry is pending sync confirmation.

8.5 Category E — Brand Management (Module 14)

Brand is a mandatory field on every job. The owner manages the brand list in settings. Brand filters apply globally across the dashboard, job lists, and all analytics. Per-brand metrics include installations completed, complaints received and resolved, revenue collected, and revisit rate.

8.6 Category F — Dealer Network (Modules 17–18)

Module 17 allows the owner to create and manage dealer accounts from the settings panel. Module 18 is the dealer portal through which dealers submit both complaint and installation jobs on behalf of their customers, manage cancellation requests, and view the full history and current status of all their submissions in read-only format. All dealer-submitted jobs flow through the standard lifecycle with no special-case processing.

8.7 Optional Module 19 — Customer Reviews and Ratings

After any job is completed or resolved, review behaviour is controlled by a mode the owner sets in the system configuration panel. There are three modes:

Off — Reviews are removed entirely from the technician's interface. This is the default if Module 19 is not included.

Optional — The technician is prompted to generate and share a review link after marking a job complete or resolved. They may dismiss the prompt with no consequences.

Mandatory — The technician must generate the review link before the job is considered fully closed on their end. The link is generated automatically and the technician is presented with a copy button to share it with the customer via WhatsApp or SMS.

In all modes, the review link is unique, job-specific, and expires after 48 hours. The customer submits their rating from their own device — the technician never interacts with the rating directly. Jobs in Mandatory mode display a "Review Pending" indicator until the customer submits or the 48-hour window expires, whichever comes first. Expiry clears the indicator automatically. Reviews are tied to the specific job, technician, and brand. Low-rated jobs (1 or 2 stars) are flagged on the owner dashboard. Average star ratings per technician feed into the performance scorecard. The owner can switch modes at any time; changes apply to new completions only and do not affect already-completed jobs.

9. Analytics

Analytics are precomputed and stored rather than calculated on every request. Whenever a job is updated — through status change, payment recording, revisit outcome, cancellation, rollback, or assignment — the system triggers updates to the relevant aggregated data. Dashboards remain fast while reflecting near-real-time data. All analytics are filterable by date range and brand, and exportable to CSV by both the owner and office staff, each scoped to the data their role can view.

9.1 Technician Performance Analytics (Module 15)

A per-technician scorecard covering both installation and complaint jobs. Key metrics include: average resolution time, complaints resolved, installations completed, payment collected, pending payment, first-visit resolution rate, revisit rate, repeat complaint rate (computed per VCID), acknowledgment response time, on-time arrival rate, average lateness, no-show count, and month-on-month job and performance trends. Cancelled jobs are excluded from resolution and completion rates but included in volume counts. If Module 19 is enabled, average customer rating is also included.

9.2 Business-Wide Analytics (Module 16)

High-level view of the full operation. Covers revenue by month, job volume by month, total jobs completed, pending payment totals, average resolution and install completion times, first-visit resolution rate, repeat complaint rate (per VCID), revisit breakdown, most common revisit reasons, business-wide on-time arrival rate by month, technician leaderboard (ranked by jobs, revenue, and first-visit rate), per-brand metrics (installs, complaints, revenue, revisit rate), complaints and installations by source (direct vs dealer by month), frequent complaint customers, and busiest days and hours.

9.3 Dealer Analytics (Part of Module 16)

A dedicated dealer reporting section within the business analytics dashboard. Metrics include: jobs submitted per dealer broken down by type (complaint vs installation), filterable by date range and brand; resolution rate per dealer; average resolution time per dealer; pending jobs per dealer; a dealer leaderboard ranked by volume and resolution rate; submissions by dealer by month; the overall direct-vs-dealer split; and dealer activity broken down by brand.

10. History and Audit Trail

All significant actions generate historical records stored as part of the job timeline. Logged events include:

- Status transitions — actor, previous status, new status, timestamp
- Undo actions — actor, status reverted, timestamp
- Office staff rollbacks — actor, previous status, reverted status, mandatory reason, timestamp
- Assignment and reassignment changes — actor, previous technician, new technician, timestamp
- Scheduling conflict acknowledgments — actor, conflicting job ID, timestamp
- Payment records and edits — actor, previous value, new value, field modified, timestamp
- Payment retain or void decisions on owner reversal — actor, decision, reason (if voided), timestamp
- Cancellation events — actor, reason, timestamp
- Cancellation request submissions, approvals, and rejections — dealer, decision maker, reason, timestamp
- Customer link actions — actor, linked customer name, timestamp
- Dismissed customer link suggestions — matched customer name, actor, timestamp
- Notes added at any stage

This provides a complete, tamper-evident chronological view of every job. Deleted records are never physically removed — they are soft-deleted, marked as deleted, and excluded from operational queries while remaining fully accessible in history.

11. Configuration Layer

A centralised settings layer, managed exclusively by the owner, controls operational thresholds and reference data used across the system. Configurable items include:

- Repeat complaint window (default: 30 days)
- Frequent complaint threshold (default: 3+ complaints per VCID in any 90-day window)
- Punctuality grace period (default: 15 minutes)
- Payment methods list (e.g. Cash, Bank Transfer, Cheque) — owner can add or deactivate
- Brand list — owner can add, rename, or deactivate
- Dealer accounts — owner can add, activate, or deactivate
- User accounts — owner creates and manages all technician and office staff accounts
- Customer review mode (Module 19 only) — owner can set to Off, Optional, or Mandatory at any time

FUTURE CONSIDERATION

The technician undo window is currently a system constant of 60 seconds. If per-client configurability is required, this value should be promoted to the Configuration Layer in a future version.

All owner actions are recorded in a system-wide activity log.

12. Technical Architecture

12.1 Backend

Built using NestJS with a modular architecture. Each domain — jobs, revisits, payments, users, brands, dealers, analytics, and settings — is isolated into its own module. Authentication is handled using JWT tokens. Role-based guards are enforced on every request. The VCID system is implemented as a backend-only identity resolution layer. The VCID identifier is never exposed in the UI; all customer linking actions are surfaced through customer-facing language (name, address, phone numbers) with the VCID resolving silently in the backend.

12.2 Frontend

Built using Next.js. Consumes the backend via a REST API. The technician-facing view is optimised for mobile use on low-end Android devices and basic browsers. The 60-second undo countdown begins immediately on the device at the moment of queue submission and is managed client-side. The server-side lock is enforced once the queued entry syncs. The technician view is configured as a Progressive Web App (PWA) to cache the app shell and static assets locally, enabling the UI to load under poor connectivity. Status transitions and payment details are queued together as a single atomic entry in the browser sync queue and retried in the background until confirmed by the server. A subtle indicator is shown if an entry is pending sync confirmation.

12.3 Infrastructure

Deployed on AWS. Backend runs on EC2 behind an Application Load Balancer (ALB), allowing additional instances to be added for redundancy without DNS or architecture changes. Database runs on RDS using PostgreSQL in Multi-AZ mode, with a synchronous standby replica in a second availability zone for automatic failover. Failover requires no manual intervention and typically completes in under two minutes.

RDS automated backups are enabled with a 14-day retention period and point-in-time recovery (PITR), allowing the database to be restored to any second within the retention window. Weekly manual snapshots are stored in a separate S3 bucket as a long-term archive outside the RDS backup cycle. The frontend can be deployed separately via CloudFront.

Three environments are maintained: Development (local machines and a shared dev database), Staging (a mirror of production used for QA, UAT, and migration validation), and Production (isolated network with strict access controls). All deployments follow the path Dev → Staging → Production. Database migrations are validated in Staging before being applied to Production. Environment variables including database credentials, JWT secrets, and API keys are managed per environment using AWS Secrets Manager and are never hardcoded or shared across environments.

12.4 Data Integrity Rules

- No job can exist without type, status, brand, and source. A technician assignment is not required at the moment of creation but is required before a job can advance beyond its initial state.
- Dealer-linked jobs must always reference a valid dealer record.
- Complaint-specific fields must not appear on installation jobs and vice versa.
- Revisits cannot exist without a parent complaint.
- Payments cannot exist without a job.

- Every job is assigned a VCID before it is saved — either linked to an existing one or newly generated.
- Brands cannot be deleted, only deactivated.
- Payment methods cannot be deleted, only deactivated.
- Cancelled jobs are retained in full and excluded from operational queues but visible in history.
- Dealer cancellation requests cannot be resubmitted after a rejection on the same job.
- Deleted records are soft-deleted (marked, not removed) and excluded from operational queries.
- When the owner reverses a Completed or Resolved status outside the undo window and a payment record exists, an explicit retain-or-void decision is required before the reversal is applied.
- For dealer-submitted installation jobs, Pending Schedule is treated as the withdrawal-eligible initial state. A dealer may withdraw directly while the job remains in Pending Schedule. Once advanced to Scheduled, the cancellation request process applies.